

Assignment =Project

Information Security

Marks Distribution

Proposal Submission=2 marks

Project deliverables=5

Presentation/viva=3

Project Summary

You are required to either conduct research on an information security topic, or attempt a development project.

In both cases, you will submit a proposal for approval. Proposal should be a 2-3 page document outlining the project goals, scope and tentative references (for research projects).

Final submission will consist of the project output (report, code etc.) and a short presentation that will be delivered in the class.

Proposals and deliverables will be submitted in google classroom. In addition, you also have to add your project details to the following google sheet.

https://docs.google.com/spreadsheets/d/1GIJTKa24DkO_HIPR2515OHifTTj5kCT4u88Lnz2du2l/edit?usp=sharing

Due dates

Proposal	8 th Nov 2025
Deliverables submission	5 th Dec 2025
Presentation	Will be shared later

Group Work

Project can be done individually or in a team of up to three students. In the latter case, every team member will also submit a statement of their own contribution to the project.

Plagiarism Policy

No two groups can choose the same project. Topics will be allocated on first come first serve basis.

Development Projects

Your aim should be to implement a prototype security product or protocol. Alternatively, you can also develop an attack or exploit toolkit. You can target a specific operating system (Windows, Linux, Android) or Web platform. Use of machine learning algorithms is encouraged. Implementation can be done in any programming language.

Some possible broad topics are given below. Your proposal **must narrow down** the chosen topic to a more specific use case. Of course, you can also come up with another topic not in the list below.

- malware and anti-malware
- intrusion detection
- end-to-end encryption
- differential privacy
- digital rights management (DRM)
- facial recognition
- DDoS attack
- digital signing
- homomorphic encryption
- firewall
- forensic analysis
- zero knowledge proof
- steganography
- IoT security
- phishing education campaign
- security automation tools
- penetration testing

Example Past Projects

- AI-based classification of phishing pages
- Creating a stealth keylogger
- Implementation of Enigma Machine
- Deep packet inspecting firewall for Android
- Malicious PDF file detection using Machine Learning

Final deliverables should include

- the project source code
- instructions on how to run your code
- documentation about the project architecture
- security analysis: discussion on your findings/results/performance/lessons.

Research Projects

For this kind of project, your aim should be to prepare a research report on a specific topic. It can be a literature review, a comparative analysis, or solution to a security problem.

Expected word count in research report is 3000-4000, excluding references.

Along with your work, you have to submit plagiarism check report. Report can be obtained from the library by sending them your final document.

Choose one specific research area (suggestions below) and look up relevant journal articles. Read their abstracts and introductions to see what specific problem is being addressed in those papers. After reviewing several papers, you will be able to narrow down the area and choose a more specific research topic. Once a sufficiently specific topic is chosen, look up more relevant research papers, and read them in detail to get a thorough understanding.

Example research areas

- Application security
- **Explainable AI**
- Cyber physical systems
- Malware analysis
- Cloud security
- Digital forensics
- Blockchain
- Infrastructure security
- Web security
- Mobile security
- Software defined firewall and industry 5.0 security
- Virtualized cyber security for cloud and fog computing
- Cryptography
- Network security protocols
- Side channel attacks detection and mitigation
- Intrusion detection and prevention
- Vulnerability management
- Security-enhanced architectures and services
- Software defined security
- Digital trust and repudiation
- Identity and trust management
- Hybrid wireless sensor network architectures and security issues
- Security aspects for a secure hardware
- Security architecture for mobile devices

Relevant journals to explore

- International Journal of Information Security
- International Journal of Security and Networks
- IEEE Transactions on Information Forensics and Security
- IEEE Transactions on Dependable and Secure Computing (TDSC)
- IEEE Security & Privacy
- New Generation Computing
- Journal of Computer Science and Technology (JCST)

Other journals: <https://research.com/journals-rankings/computer-science/computer-security-cryptography>

Where to search for journal papers

Instead of google, you will find more relevant articles on research databases like below. When doing a search, **make sure you filter the results for 'journal' articles**, leaving aside conference papers and books.

[IEEE Xplore](#)

[Science Direct](#)

[ACM Portal](#)

More databases here: [NUCES Digital Library](#)

Example Past Projects

- Vulnerability Analysis: Remote Code Execution
- Threats of banking trojans and its prevention
- Comparative analysis of ransomware detection techniques in android devices
- Using honeypots for intrusion detection and prevention in cloud environments
- An analysis of Elliptic Curve Cryptography and its applications
- Vulnerability analysis for different multiplayer solutions for Unity game engine
- Social Engineering attacks in South Asian countries and their prevention measures
- Steganography in modern times

Final Report Structure

A suggested structure is below. Some variation might be needed based on your topic.

1. Abstract

Provide an executive summary of the whole report. 150 words on average.

2. Introduction

Describe the background of the topic and the research goals.

3. Motivations

Tell the reader why this topic is important.

4. Methodologies (or Literature Review)

Discuss all the methodologies that you have compared, under separate subheadings

5. Analysis

Compare all the previous methodologies. Try to have a critical view of the whole situation. If possible, identify any room for further research work.

6. Conclusion

Summarize the main findings, provide recommendations (if any).

7. Reference List

List all references in IEEE style. Make sure you cite all references within the text.